

Given that the frequency of knowledge in training datasets significantly influences a model’s ability to retain and comprehend it (Allen-Zhu & Li, 2023; Meng et al., 2022; Mallen et al., 2023), we utilize Wikipedia page views as a proxy for knowledge frequency in the models’ pre-training datasets[¶]. To this end, we calculated the page views for each concept on Wikipedia between January 1, 2010, and December 31, 2019[‡]. Based on these statistics, concepts are categorized by page view frequency into three equal tiers: low-frequency (bottom 33% of the distribution), medium-frequency (middle 33%), and high-frequency (top 33%). A more detailed distribution of the categories and the corresponding example data of SpecWiki dataset are provided in Table 4 and Table 5, respectively, in the Appendix. This approximation helps estimate the likelihood of a concept’s presence in the models’ pre-training datasets and allows us to explore how knowledge at different frequency levels is stored in models and provides a more comprehensive evaluation.

Question Generation To more precisely assess the retention of knowledge within the model, we design two sets of question formats.

- *Multi-Choice Questions.* Drawing inspiration from the widely used Massive Multitask Language Understanding (MMLU) dataset (Hendrycks et al., 2021b), which evaluates general knowledge across models, we similarly designed ten multiple-choice questions for each concept, ensuring the knowledge and answers could be directly found in the relevant Wikipedia articles. Specifically, we provided GPT-4o (OpenAI et al., 2024) with the appropriate Wikipedia article for each concept and instructed it to extract ten questions without overlap, along with the correct answers derived from the article’s text. Next, it was instructed to generate three additional incorrect answers, aside from the golden answer, ensuring that none of them overlapped with the correct answer to avoid confusion. The detailed prompt is available in §A.1. We also include sample multiple-choice questions and results of manual verification of the generated data in Appendix §A.2.
- *Open-ended Generation.* To more effectively assess the model’s ability to generate knowledge text freely, and to overcome the randomness and lack of depth inherent in the Multi-Choice Question evaluation method, we also set up a series of Open-ended Generation questions. For each question related to a concept, we prompted the model to generate an answer of up to 150 tokens directly and used GPT-4o as an evaluator to evaluate whether the generated response correctly matched the golden answer.

4 Experiment

4.1 Experimental Setup

Evaluated Models To provide a more comprehensive evaluation of how the degree of parameter specialization evolves across large language models, we assessed 20 open-source models from various families and sizes in the community. Specifically, we evaluated LLaMA series (Touvron et al., 2023a,b; Grattafiori et al., 2024), Qwen series (Bai et al., 2023; Yang et al., 2024; Team, 2025), Gemma series (Team et al., 2024a,b), OLMo series (Groeneveld et al., 2024; OLMo et al., 2025), Yi series (AI et al., 2025), Mistral series (Jiang et al., 2023), GPT-j-6b (Wang & Komatsuzaki, 2021), Pythia-6.9b (Biderman et al., 2023), Falcon-7b (Almazrouei et al., 2023) and Mpt-7b (Databricks, 2023). Refer to Appendix §B.1 for the implementation details of these models.

Knowledge Vectors Masking Setup Based on the descriptions in §3.2, in order to obtain $\mathbf{m}^\ell$ and $\mathbf{m}^{*\ell}$ for each concept in SpecWiki at each layer of the model, we set the number of concept-related questions t to 10. Additionally, we randomly select 5 irrelevant concepts with no knowledge overlap from the benchmark, and gather the corresponding questions associated with these irrelevant concepts, resulting in $t^* = 50$ irrelevant questions. These collected questions will also be directly utilized in the computation of both the Concept-Specific Score and the General Score.

[¶]To better support this point, we include experiments in §A.4 of Appendix that validate the strong correlation between concept popularity and their frequency in the pretraining data.

[‡]The earliest release date of the evaluated models, such as GPT-J, is 2019. Therefore, their pretraining datasets could not include knowledge or concepts that emerged after this period. To ensure a fair evaluation across all models, any knowledge introduced post-2019, including the COVID-19 pandemic, was excluded from the benchmark.



Figure 2: Correlation between the performance on SpecWiki and parameter specialization score (PSS) in 20 language models. We use a color gradient to distinguish the release times of the models, with cooler colors indicating earlier release dates and warmer colors representing later releases. Additionally, the size of each circle reflects the model’s performance on MMLU, with larger circles indicating better performance. The blue trendline, obtained through linear regression fitting of the data points, suggests a strong correlation between a model’s performance on SpecWiki and its degree of Parameter Specialization.

Regarding the selection of model layers for masking, since the initial layers of a model typically handle fundamental capabilities like basic text processing (Meng et al., 2022; Geva et al., 2023), masking these layers could severely impair the model’s basic text generation abilities. Therefore, for all models in our study, we preserve the first 5 layers without masking and only apply vector masking operations to all subsequent layers.

For the PSS computation of each model, we selected five different fixed k values—10%, 20%, 30%, 40%, and 50%—which represent the proportion of value vectors in the model’s MLP layers that were masked. For each k , we calculated the corresponding PSS following 6, and then averaged the results to obtain the final PSS score for each model. This criterion was applied consistently across both the Multiple-Choice Questions (MCQ) and Open-ended Generation (OEG) tasks.

4.2 Main Results

The main results for the Multiple-Choice Questions setting can be seen in Figure 2. We observe a strong correlation between the degree of Parameter Specialization (measured by PSS) and model performance on SpecWiki across 20 models, with Pearson and Spearman coefficients of 0.92 and 0.93, respectively. Models achieving better performance on SpecWiki exhibit higher Parameter Specialization Scores. Furthermore, models with higher PSS are often those released more recently (warmer color) and exhibit stronger general abilities, as measured by their MMLU performance (larger circle). The corresponding results for the Open-ended Generation setting can be found in Figure 6 in §B.2, which exhibit similar patterns and trends.

To better analyze the variations in Parameter Specialization across models within the same family, we selected eight models from four model families: LLaMA, Qwen, Mistral, and Gemma. We examined how the difference between the General Score, which represents the model’s ability to handle irrelevant knowledge, and the Concept Specific Score, which reflects the model’s ability to handle task-specific knowledge, changes under different masking ratios of parameter vectors. The results are shown in Figure 3.

From the figure, we can observe a very similar pattern across models from the four families:

1. Among models within the same family, more advanced models tend to achieve higher peaks in the General Score - Concept Specific Score difference. This indicates that more advanced models generally exhibit higher levels of Parameter Specialization.

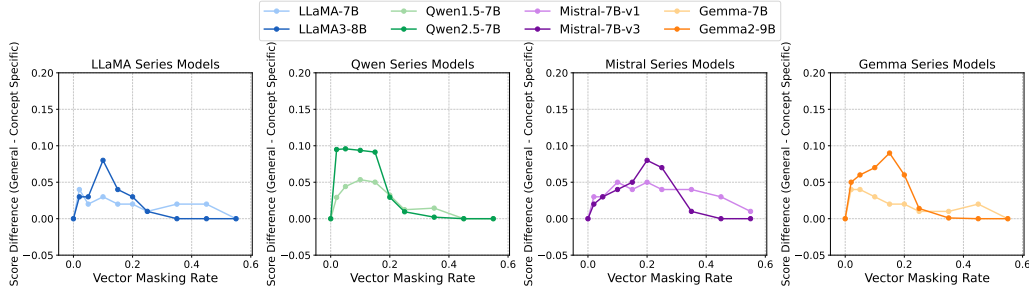


Figure 3: Analysis of Parameter Specialization variations across models within the same family. We selected eight models from four model families: LLaMA, Qwen, Mistral, and Gemma. The figure shows how the difference between the General Score (representing the model’s ability to handle irrelevant knowledge) and the Concept Specific Score (representing the model’s ability to handle task-specific knowledge) changes under different masking ratios of parameter vectors.

- As the masking ratio of parameter vectors increases, from approximately 5% to 20%, the difference between the General Score and the Concept Specific Score gradually increases to a peak. This indicates that we are removing parameter vectors that are highly specific to the target knowledge. After reaching the peak, as the masking ratio continues to increase, the difference gradually decreases to zero. This suggests that parameter vectors with lower activation are often those that have a higher degree of knowledge superposition and are less specialized in the target knowledge.

Additionally, we unexpectedly found that when a small proportion of concept-related vectors (ranging from 5% to 10%) were masked, the performance of the masked models on unrelated questions even surpassed that of the original models. This observation is consistent across various models and indicates the positive impact of reducing irrelevant information interference in the model’s representation, leading to improved performance.

In §5, we will further validate the causal relationship between the degree of model parameter specialization and its ability to better utilize target knowledge through the finetuning experiments on additional data.

4.3 Impact of Model Scale on Parameter Specialization

In this section, to better explore the differences in the degree of parameter specialization across models of different sizes, we conducted Knowledge Vectors Masking experiments on five Qwen1.5 models of varying sizes (0.5B, 1.8B, 4B, 7B, and 14B) and on 2 Gemma2 models (2B and 9B). The results are shown in Table 1. We observe that in both the Qwen and Gemma model families, as the model size increases, the corresponding Parameter Specialization Score also increases. This trend is accompanied by improved performance on SpecWiki. This suggests that in larger-scale models, the degree of superposition for specific knowledge decreases and it tends to be distinctly represented across designated parameter vectors.

Model	Accuracy _{MCQ} ↑	PSS ↑
Qwen1.5-0.5B	0.61 (± 0.2)	0.019 (± 0.01)
Qwen1.5-1.8B	0.61 (± 0.3)	0.044 (± 0.02)
Qwen1.5-4B	0.73 (± 0.2)	0.075 (± 0.02)
Qwen1.5-7B	0.75 (± 0.2)	0.121 (± 0.04)
Qwen1.5-14B	0.82 (± 0.2)	0.184 (± 0.03)
Gemma2-2B	0.72 (± 0.3)	0.057 (± 0.02)
Gemma2-9B	0.86 (± 0.1)	0.138 (± 0.03)

Table 1: Performance comparison of language models with varying sizes on Multiple-Choice Question and Parameter Specialization Score. Both the Qwen1.5 and Gemma2 series models show improved Parameter Specialization as the model scale increases, accompanied by better performance on the MCQ testing in SpceWiki.

4.4 Evolution of Parameter Specialization During Pretraining

To better investigate the development of Parameter Specialization in the model from the perspective of model training dynamics, we analyzed 10 checkpoints from the OLMo-2-1124-7B (OLMo et al., 2025) pretraining process by using our SpecWiki. The results are shown in Figure 4 below.

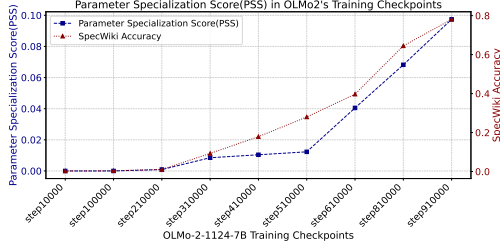


Figure 4: Development of Parameter Specialization in OLMo-2-1124-7B over the pretraining process.

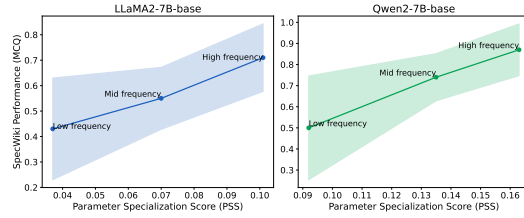


Figure 5: Relationship between concept popularity, model accuracy on MCQ, and Parameter Specialization Score in LLaMA2-7B and Qwen2-7B models.

Model	Accuracy _{MCQ} ↑	Accuracy _{OEG} ↑	PSS ↑	Semantic Entropy ↓	Local Intrinsic Dimension ↓
LLaMA2-7B	0.60 (±0.2)	0.51 (±0.1)	0.67 (±0.1)	0.67 (±0.1)	11.23 (±2.1)
LLaMA2-7B _{FT-FV}	0.63 (±0.3)	0.54 (±0.2)	0.65 (±0.2)	0.62 (±0.1)	11.12 (±1.4)
LLaMA2-7B _{FT-PV}	0.67 (±0.3)	0.59 (±0.2)	0.72 (±0.1)	0.50 (±0.2)	7.89 (±2.1)
LLaMA2-7B _{FT-CV}	0.62 (±0.1)	0.51 (±0.1)	0.63 (±0.2)	0.62 (±0.1)	11.12 (±1.4)
LLaMA2-7B _{FT-RV}	0.58 (±0.2)	0.49 (±0.2)	0.65 (±0.2)	0.65 (±0.2)	11.07 (±2.7)
Qwen2-7B	0.72 (±0.3)	0.63 (±0.1)	0.124 (±0.03)	0.56 (±0.1)	9.78 (±1.9)
Qwen2-7B _{FT-FV}	0.73 (±0.2)	0.67 (±0.2)	0.110 (±0.02)	0.59 (±0.2)	8.53 (±1.1)
Qwen2-7B _{FT-PV}	0.77 (±0.1)	0.70 (±0.1)	0.133 (±0.03)	0.39 (±0.1)	6.92 (±1.3)
Qwen2-7B _{FT-CV}	0.73 (±0.2)	0.65 (±0.2)	0.114 (±0.03)	0.55 (±0.2)	8.78 (±1.6)
Qwen2-7B _{FT-RV}	0.71 (±0.2)	0.63 (±0.1)	0.122 (±0.02)	0.59 (±0.2)	9.65 (±1.4)

Table 2: The performance of both the original LLaMA2-7B-base and Qwen2-7B-base models, along with their FT-FV, FT-PV, FT-CV and FT-RV variants, was assessed on a selection of 10 high-frequency concepts from SpecWiki. Five metrics were used to evaluate their performance, including their general effectiveness, Parameter Specialization, and the degree of hallucination present in their output.

From the results, we observe that during the early training steps (step 10,000 to step 210,000), both the PSS and the accuracy on SpecWiki remain nearly unchanged and close to zero. In the subsequent phase (step 310,000 to step 510,000), although the model begins to show noticeable gains in accuracy on SpecWiki, the PSS are still under 0.1. However, it is during the later training steps (step 610,000 to step 910,000) that parameter specialization begins to emerge, accompanied by a more substantial improvement in accuracy. These findings suggest that parameter specialization does not occur in the early stages of training, but rather emerges after a certain amount of data exposure. Furthermore, as training continues and the model sees more data, the degree of parameter specialization increases accordingly.

4.5 Parameter Specialization in Relation to Concept Popularity

In this section, we analyze how the popularity of concepts themselves, which is roughly equivalent to their frequency in the pretraining data, will affect the level of parameter specialization for the corresponding knowledge. We follow the classification method for concepts as outlined in §3.4, dividing them into high-frequency, mid-frequency, and low-frequency categories. The impact on their PSS scores is measured on two example models, LLaMA2-7B, and Qwen2.5-7B, which are shown in Figure 5.

From the figure, it is clear that in both the LLaMA2-7B and Qwen2-7B models, as the popularity of a concept decreases, the model’s accuracy on that specific knowledge declines, accompanied by a lower Parameter Specialization Score. This suggests that the degree of Parameter Specialization for a particular knowledge in the model’s parameters is likely directly correlated with the frequency of that knowledge in the model’s pretraining dataset. The higher the frequency, the greater the Parameter Specialization for that knowledge in the model.

5 Validation of Parameter Specialization Benefits for Knowledge Tasks

In this section, we conducted four sets of controlled training experiments, each involving continued fine-tuning on the Llama2-7B-base (Touvron et al., 2023a) and Qwen2-7B-base (Yang et al., 2024)